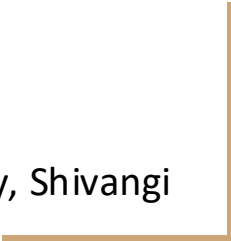


TBI on a BBB Chip to study the effects of Curcumin

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Sarkar



Introduction

Traumatic Brain Injury

- Nondegenerative and Noncongenital insult to the brain caused by external force
- Occurs from violent blow or jolt to the brain
- permanent / temporary impairment of cognitive, physical, and psychosocial functions

Primary Injury

- Injuries that occur immediately after the incident
- Sudden or profound injury
- Caused by direct mechanical forces exerted on the brain

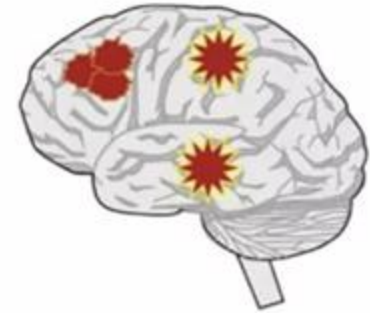


Primary Injury

**Tissue deformation, axonal shearing,
contusion, necrosis, blood-brain barrier disruption**

Secondary Injury

- Hours to days after injury primary injury
- Reactive process after the initial head trauma
- series of steps of cellular, chemical, tissue, or blood vessel changes which may cause destruction of brain tissue.
 - **Chemical, molecular, and inflammatory** effects take place in the brain



Secondary injury

Due to:

Increased intracranial pressure

Hypoxia

Hypotension

Hypothermia

Electrolyte disturbances (Na, Ca, K)

Toxic amino acids

Oxygen radicals

Introduction: Blood Brain Barrier (BBB)

selective semi-permeable membrane between the blood and the interstitium of the brain

Consists of...

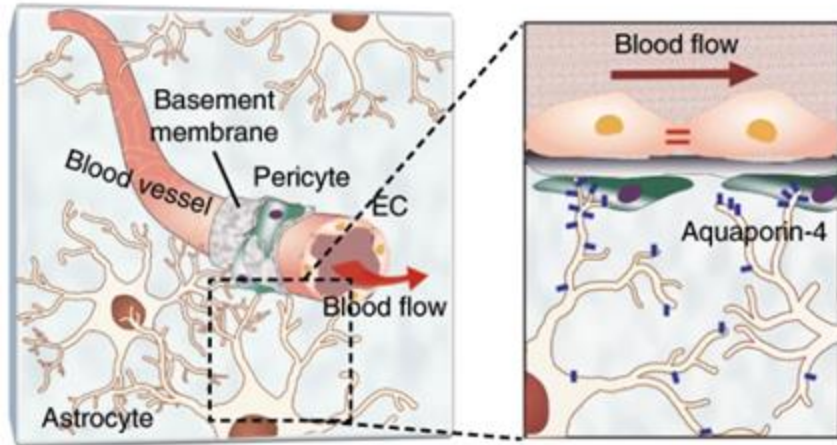
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astrocy

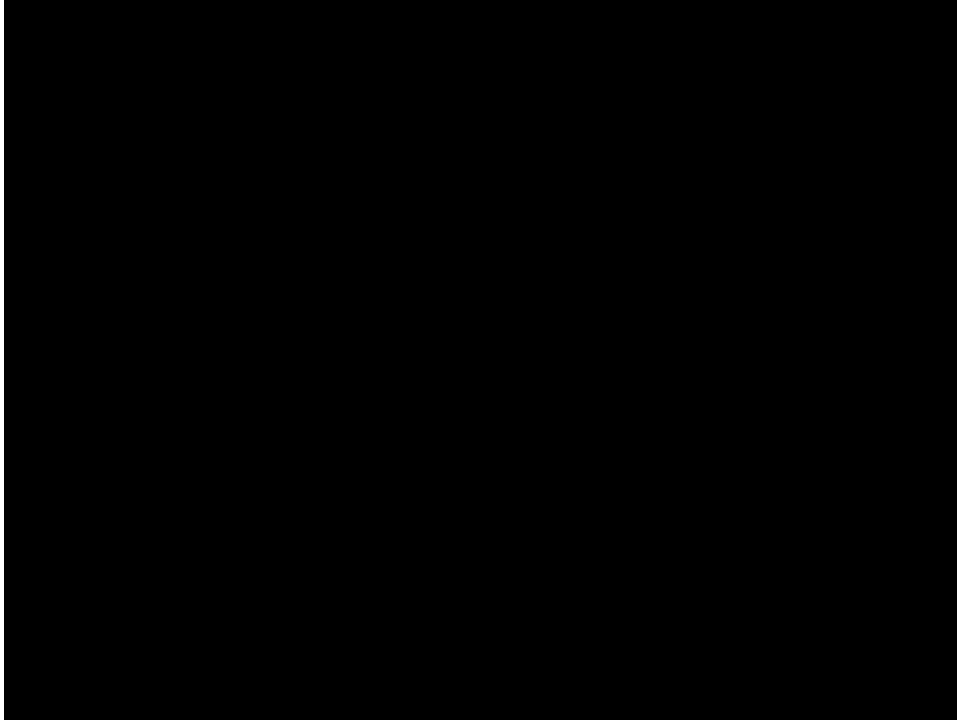


<https://doi.org/10.1038/s41467-019-13896-7>

Introduction: purpose of the BBB

1. Selective permeability
2. Neuroprotection
3. Brain homeostasis
4. Neurotransmitter levels

More detailed view on of IBI effects on BBB



Steps of the destruction of BBB

1. Macrophages are activated
2. Cytokines are released
3. Abundance of adhesion molecules → creation of binding sites

Introduction: TBI effects on BBB

- **Shearing forces** leads to breaches in the BBB
- **Inflammatory responses** can disrupt the tight junctions of endothelial cells
- **Oxidative stress** can impair the BBB integrity and function
 - Occludin and claudin
- **Changes in neurotransmitter level** can affect the BBB permeability
- Decreased TEER
- Increased permeability

TBI: Chemical components

Oxidative Stress

Mediators: ROS and antioxidant imbalance

Mitochondrial Damage

Mediators: Excess intracellular calcium ions and electron transport chain uncoupling

Excitotoxicity

Mediator: excess glutamate



Neuroinflammation

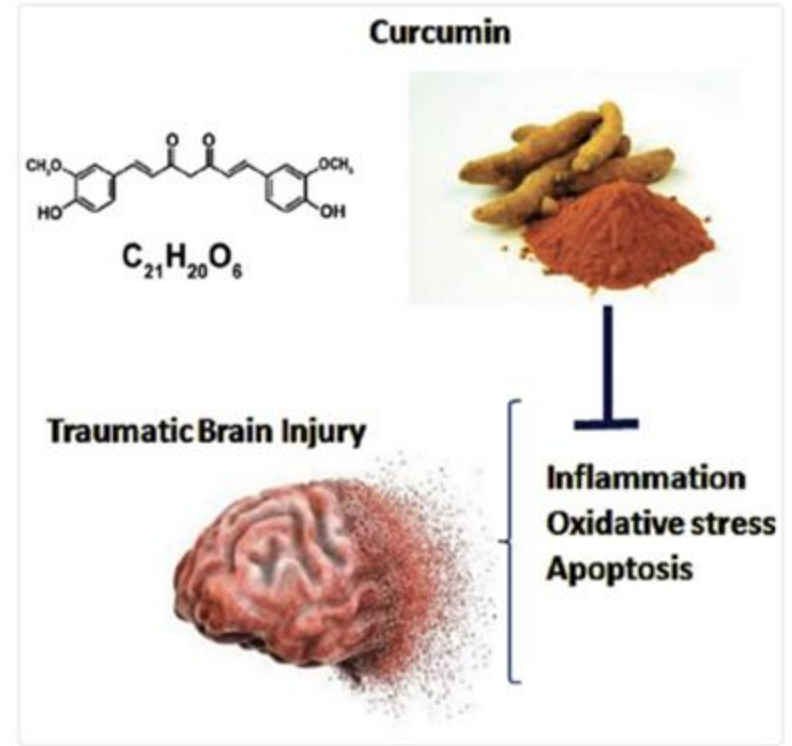
Mediator: $\text{TNF-}\alpha$, $\text{IL-1}\beta$, $\text{IFN-}\gamma$, nitric oxide, and ROS

Cell Death

Mediator: Caspases, cysteine proteases, and free radicals

Curcumin treatment

- Curcumin is a dietary antioxidant found in turmeric, known for its potential health benefits.
- Experimental studies have shown that curcumin can help improve outcomes in TBI by reducing damage caused by oxidative stress.
- Curcumin shows promise as a therapeutic agent for TBI due to its antioxidant, anti-inflammatory, and neuroprotective properties.
- It acts on multiple pathways involved in TBI pathology, including oxidative stress, inflammation, and neuronal survival.
- Further research is needed to fully understand the mechanisms of curcumin's effects and its potential for clinical use in TBI treatment.



Current Methods

In Vivo Testing of Curcumin

Study Design	Independent Variable	Control	Treatment Duration	Results
Male Rats	Cur (10 mg/kg, i.p.); Cur (20 mg/kg, i.p.); Cur (30 mg/kg, i.p.); Cur (50 mg/kg, i.p.);	Vehicle (Saline)	5 weeks	• ↓ Levels of NLRP3, IL-1β, IL-6, IL-18, TNF-α • ↑ BDNF/TrkB and PI3K/Akt signaling pathways

Current methods

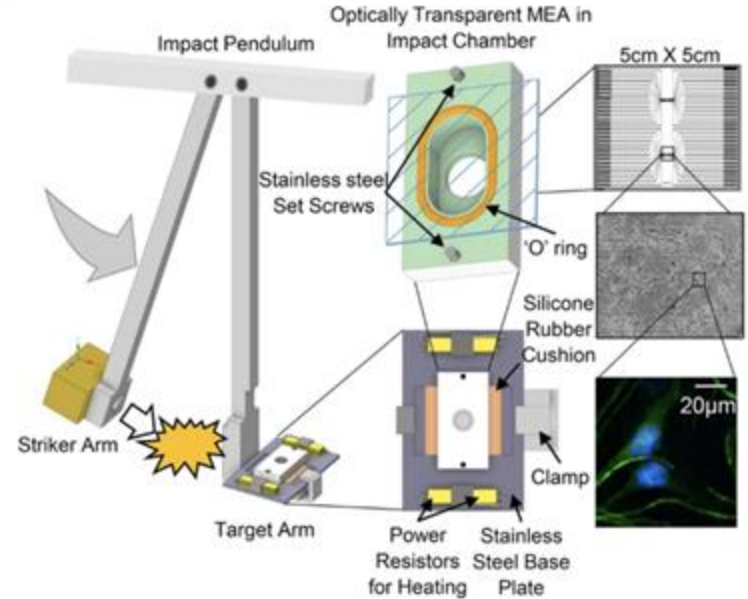
Primary Neuron Culture

1. Primary Neuron Cultures were taken from frontal cortex tissues taken from embryos of mice
2. Placed into microelectrode array after inserted with couple of serum

Pendulum Model

1. Impact pendulum swings causing striker arm to hit the target arm

Figure 1



<https://doi.org/10.1038/s41598-022-14937-w>

These impact pendulum tests reveal that force dependent factors can cause an increase in inflammation and oxidative stress

Current methods: Mimicking TBI on Blood Brain Barrier In Vitro

Wall Shear Stress

In order to create a shear stress equivalent EGM plus media was perfused through a peristaltic pump

TNF - α

Study effects of permeability and tear

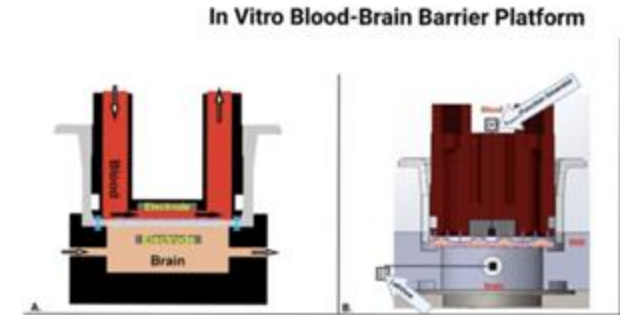
OGD/R

Oxygen glucose depleted environment

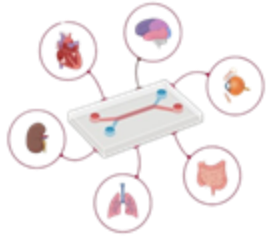
Glucose free DMEM allows for OGD conditioning- hypoxic environment

Dextran Permeability Assay Test

Permeability coefficient calculated based on concentration of dextran, observation period and surface area



Objective & Research Question



Objective: Develop a TBI-on-a-chip platform replicating the blood brain barrier (BBB) microenvironment after injury

Research Question: Developing a TBI on a chip platform replicating the BBB to study drug testing
- curcumin



Device Design and Fabrication

Materials Used:

- Polydimethylsiloxane (PDMS) for casting channels.
- Custom 3D printed mask molds for shaping PDMS.

Channel Specifications:

- Channels were 2 cm long and 1 mm wide.
- Top channel was 1 mm high, bottom channel was 0.2 mm high.

PDMS Casting:

- PDMS parts were cast using a 10:1 ratio of base to curing agent.
- Degassed at -80 kPa to remove bubbles.
- Cured at 60 °C for at least 4 hours.

Inlet and outlet formations

- Biopsy punches of 1 mm made

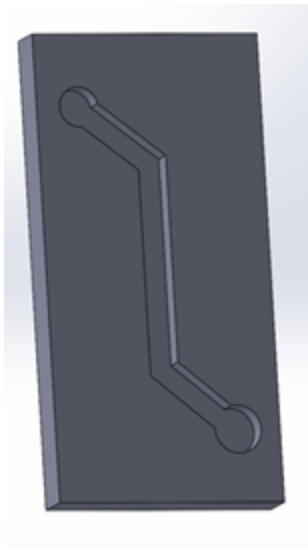
Membrane Preparation:

- Epoxy silane method used for bonding membranes to PDMS.
- A polyethylene ter-phthalate (PET) membranes with 0.4- μ m perpendicular pores
- Parts and membranes treated with O₂ plasma and submerged in specific silane solutions for bonding.
- Parts aligned and bonded by hand, then placed in a 60 °C oven under weights for at least 12 hours for annealing.

Chip Layers

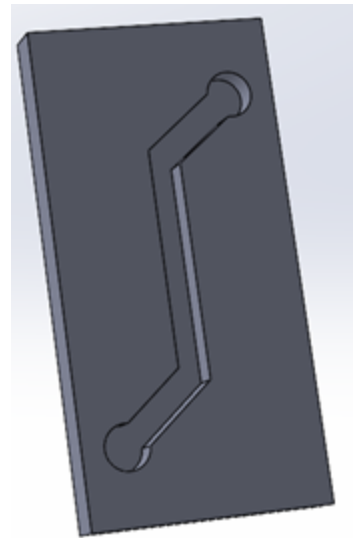
Top Layer

- Containing astrocytes and pericyte network



Bottom Layer

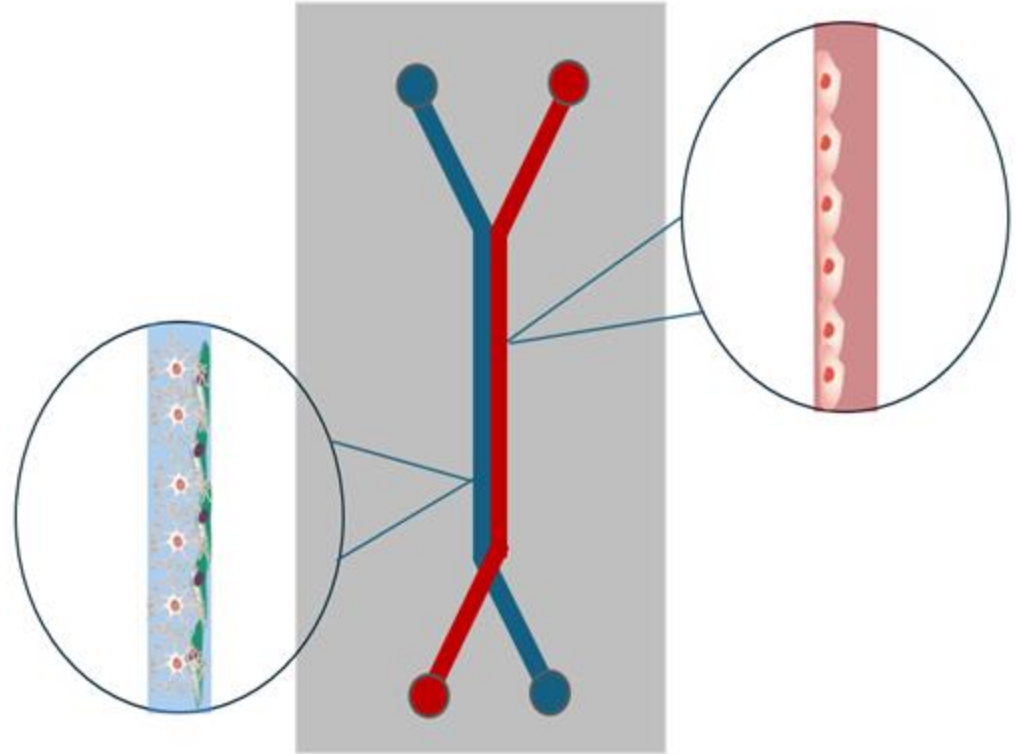
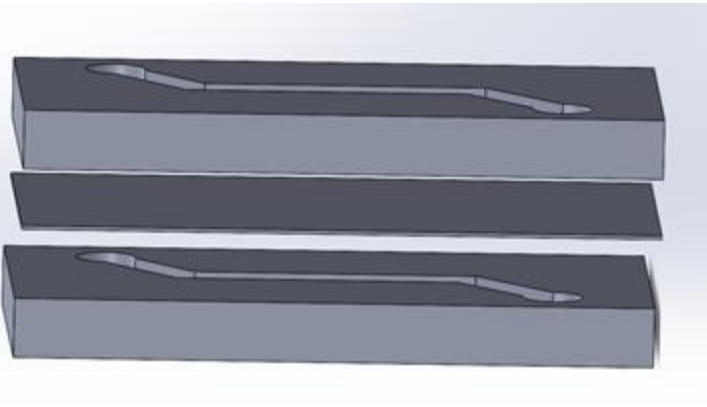
- Containing endothelial monolayer



Device Assembly

Side view:

- Top and bottom layer connected with porous membrane



Cell Seeding

1. Microfluidic Device Design:

- The upper channel represents the central nervous system (CNS) microenvironment, while the lower channel represents the vascular system.
- The channels are separated by a porous membrane coated with extracellular matrix (ECM) proteins (collagen type IV and fibronectin).

2. Cell Seeding and Culture:

- A mixture of primary human brain astrocytes and pericytes, in a ratio of 7:3, were cultured on the upper side of the porous membrane in the upper channel for 1 hour.
- Human induced pluripotent stem cell-derived brain microvascular endothelial cells (iPS-BMVECs) were cultured on the bottom side of the porous membrane in the lower channel.
- Chip was cultured for one additional day to promote endothelial monolayer formation and to acclimate the cells to the device

3. Adhesion and Monolayer Formation:

- iPS-BMV ECs were allowed to adhere to the membrane by incubating the microfluidic chip upside down for 5 hours.

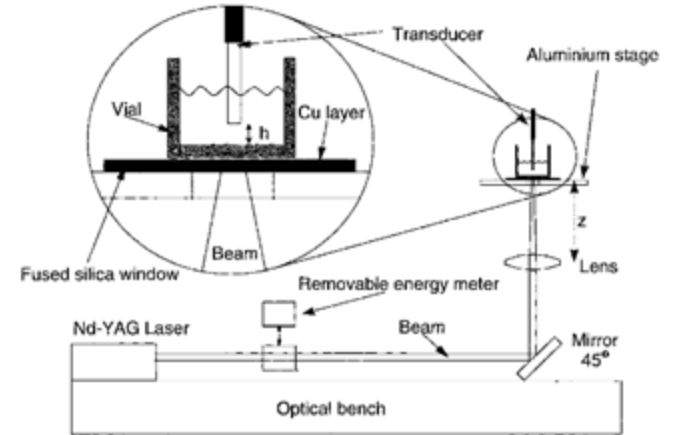
4. Flow Control and Shear Stress:

- Flow through the vascular channel was controlled by microfluidic pumps to maintain physiological levels of fluid shear stress (6 dyne cm^{-2}) and blood-like viscosity (3–4 cP).

Methods

Physical Injury : Shock wave induced injury

- Take the chip and put in a shock chamber
- Laser induced shock wave (LIS method)
- Shock waves administered in the chamber



[https://doi.org/10.1002/\(sici\)1096-9101\(2000\)26:4%3C364::aid-lsm4%3E3.0.co;2-t](https://doi.org/10.1002/(sici)1096-9101(2000)26:4%3C364::aid-lsm4%3E3.0.co;2-t)

Methods

Teer Measurement

1. Prepare the Setup:

- The chip should be positioned such that the electrodes of the measurement system are in contact with the medium in both the upper CNS microchannel and the lower vascular microchannel.

1. Electrical Impedance Measurement:

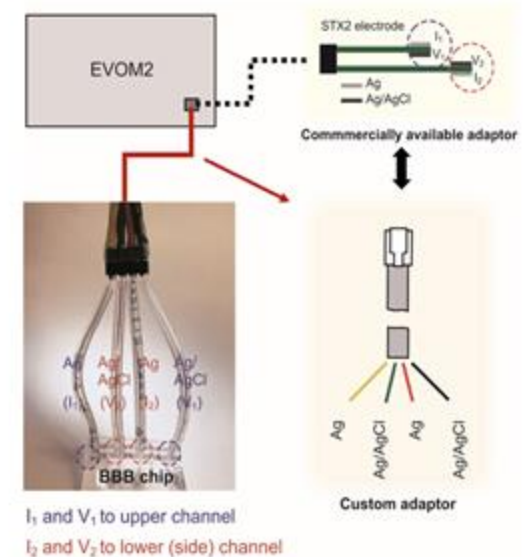
- Apply a low-frequency alternating current (AC) voltage across the electrodes.
- Measure the resulting current passing through the medium in the microchannels.
- The impedance (Z) of the system can be calculated using Ohm's law: $Z = V/I$
- The Ag/AgCl electrode integration directly into the platform

3. Calculate TEER:

- TEER is calculated using the formula: $TEER = (Z_{\text{sample}} - Z_{\text{blank}}) \times \text{Area}$, where Z_{sample} is the impedance measured across the cell monolayer, Z_{blank} is the impedance measured across a blank area of the chip (without cells), and Area is the cross-sectional area of the membrane.

1. Data Analysis:

- Record the TEER values at different time points..



Each electrode wire is 3 cm long and placed in a Tygon tubing filled with culture medium for cell culture.

Methods

Permeability measurement

Permeability Assay:

- 500 $\mu\text{g mL}^{-1}$ FITC-dextran introduced into the lower channel.
- Incubate and collect samples from both upper and lower channels.

Assay Setup:

- Use of fluorescence microplate reader or microscope for measurement.
- Calculate permeability coefficient using collected data.
- $P = (dQ/dt) / (A \times C_0)$

Data Analysis:

- Assess permeability of endothelial monolayer.
- Compare permeability under different conditions.

Methods

Introduction of secondary injury

- Preparing secondary injury factors NLRP3 (1-10 ng/ml), IL-1 β , IL-6, IL-18, and TNF- α (1-100 ng/ml) solutions at specific concentrations.
- Administering the secondary injury factors into the medium flowing through the lower vascular microchannel of the BBB chip at 22 hours after the introduction of primary injury.
- Using a microfluidic pump to control the flow rate and volume of the solution containing the secondary injury factors.
- Allowing the secondary injury factors to flow through the BBB chip for 6 hours of incubation period.
- Measuring changes in transendothelial electrical resistance (TEER) to assess barrier integrity.
- Quantifying the levels of inflammatory cytokines (IL-1 β , IL-6, IL-18, TNF- α) using enzyme-linked immunosorbent assay (ELISA)

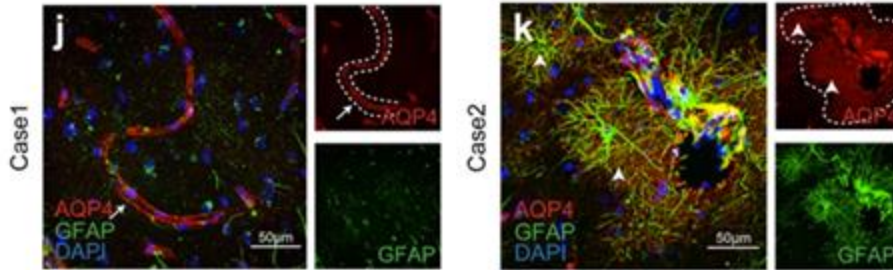
Methods

Administration of Curcumin:

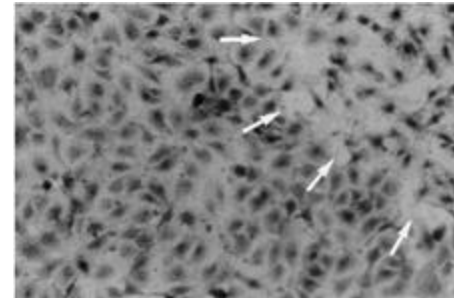
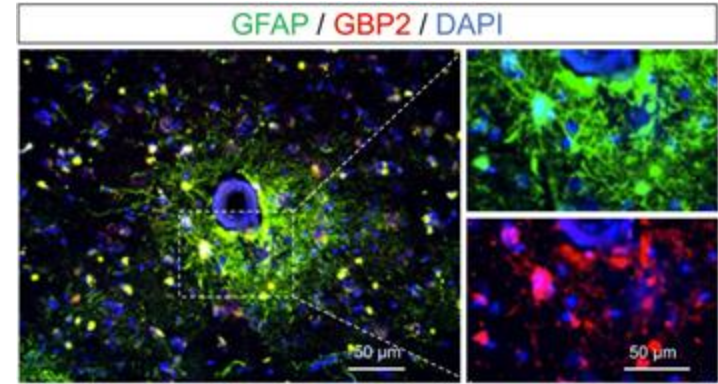
- Preparing the curcumin solution in appropriate DMSO solvent to prepare a stock solution at 28 hrs.
- Administering Curcumin directly into the medium flowing through the lower vascular microchannel of the BBB chip.
- Using a microfluidic pump to control the flow rate and volume of Curcumin solution administered.
- Allowing the Curcumin-treated medium to flow through the BBB chip for 5 weeks to ensure exposure of the endothelial monolayer to the compound.
- Control treatment : untreated sample of BBB chip to assess the specificity of Curcumin effects and also comparing with previous in vivo studies.
- Evaluating gene expression changes using qPCR

Results Expectation: Modeling TBI

- Disruption of BB tight junctions
- Upregulation of reactive astrocytes
 - **GFAP**
 - **GBP2**
- Upregulation of **AQP4**



<https://doi.org/10.1038/s41598-017-02505-6>



[https://doi.org/10.1002/\(SICI\)1096-9101\(2000\)26:4<364::AID-LSM4>3.0.CO;2-](https://doi.org/10.1002/(SICI)1096-9101(2000)26:4<364::AID-LSM4>3.0.CO;2-)

Results Expectation: Curcumin Treatment

Gene expression analysis using **qPCR**:

- A decrease in the expression of pro-inflammatory genes such as IL-1 β , IL-6, TNF- α , and COX-2. This downregulation indicates a reduction in neuroinflammation following Curcumin treatment.
- An upregulation of anti-inflammatory genes such as IL-10 and TGF- β . This indicates a shift towards an anti-inflammatory environment in response to Curcumin treatment.
- Downregulation of oxidative stress-related genes such as Nrf2 and HO-1, indicating a decrease in oxidative damage.
- Changes in the expression of NF- κ B and MAPK signaling pathways, indicating normalization of inflammatory signaling.



Research Impact

- Currently, no TBI-on-a-chip model which replicates both **primary** and **secondary** injury
- Introduction of a **no-contact** physical injury of TBI
- No therapeutic for TBI
 - Utilizing a natural herbal treatment



Future work

- Extending the application of the BBB chip device beyond TBI to study other neurological disorders characterized by BBB dysfunction, such as stroke, Alzheimer's disease, and multiple sclerosis.
- Utilizing induced pluripotent stem cells (iPSCs) derived from patient samples to create personalized BBB chip models of TBI. This would allow researchers to study individual variability in drug responses and identify personalized treatment strategies.
- Further diversify the cell types present in the BBB chip model to better mimic the complexity of the neurovascular unit.
- Overall, the future directions for the BBB chip device for studying TBI are focused on enhancing its physiological relevance, scalability, and predictive value, with the ultimate goal of improving our understanding of TBI pathology and developing effective treatments for patients.

Limitations

- Utilizing **LIS method** may be challenging at a microscale level
 - Inducing physical injury such as a violent blow might be difficult to execute on such a small scale and jeopardize device integrity
- Maintaining the viability and functionality of cells within the BBB chip model over long-term culture periods may be challenging and could affect the relevance of results for studying chronic aspects of TBI pathology.
- The BBB chip model typically relies on static culture conditions, which may not fully mimic the dynamic microenvironment of the brain following TBI, including changes in blood flow, shear stress, and interstitial fluid dynamics.
- While the BBB chip model allows for the co-culture of different cell types, it may still lack the full complexity of the neurovascular unit and the surrounding brain tissue.

Educational outreach



- Hosting public events, such as science fairs, open houses, or health expos, to showcase the BBB chip model and educate the community about TBI and Curcumin therapy.
- Collaborating with schools, medical institutions, and universities to incorporate the BBB chip model into their science curriculum or research projects.
- Offering workshops or seminars for students and educators to learn about the BBB chip model, TBI, and Curcumin therapy.
- Utilizing social media platforms and online channels to share educational content, updates, and resources related to the BBB chip model and Curcumin therapy.
- Engaging with followers, answer questions, and facilitate discussions about TBI research and drug testing.



Summary

- The BBB chip model provides a physiologically relevant platform for studying the effects of Curcumin on the neurovascular unit in TBI.
- Through the BBB chip model, we can investigate Curcumin's impact on BBB integrity, inflammatory responses, oxidative stress, and neuronal function in a controlled microenvironment.
- Understanding how Curcumin modulates BBB function and neuroinflammatory pathways can inform the development of novel treatment strategies for TBI.
- Continued investigation into Curcumin's therapeutic potential in TBI, leveraging innovative technologies like the BBB chip model, holds promise for improving outcomes and quality of life for individuals affected by TBI.

